Name: _____ Mark _____/45

Acids, Bases and Equations Assessment

1. In the following list of properties, tick whether you think it is the property of an acid or a base. (7 marks)

Property	Acid	Base
Sour taste		
Produces hydrogen ions in solution		
Turns litmus blue		
Produces hydroxide ions in solution		
Slippery feel		
Bitter taste		
Turns litmus red		

2. Jodie accidentally spilled oven cleaner on her hand and it started to itch and burn. She read on the bottle that it contains a strong base.

    a. Name three things in the house that she could put on her hand to stop the burning. (3 marks)

    b. Explain why these would stop the burning. (2 marks)

3. Hydrochloric acid is added to magnesium metal. This produces hydrogen gas and a salt (magnesium chloride). (3 marks)

    a. What are the:

    Reactants _____

    Products _____

    b. Write the word equation for the reaction.

    _____

4. *Azalea* plants need acidic soils to grow well and produce the best flowers. Describe the way in which you could test the soil around some *Azalea* plants to check if the soil is acidic. Be sure to say how you will be able to tell if it is acidic or not. (4 marks)

5. What information does the chemical equation below give you that the word equation does not? (2 marks)

Magnesium + Oxygen gas → Magnesium oxide

$2Mg + O_2 \rightarrow 2MgO$

6. In the pop test, hydrogen reacts violently with oxygen gas producing water.

$H_2 + O_2 \rightarrow H_2O$

   a. Is this equation balanced? Explain why. (2 marks)

   b. Explain where each of the atoms in the water molecule have come from. (2 marks)

7. Balance the following equations: (3 marks)

   a. $Mg + HCl \rightarrow MgCl_2 + H_2$

   b. $N_2 + H_2 \rightarrow NH_3$

   c. $H_3PO_4 + Zn(OH)_2 \rightarrow Zn_3(PO_4)_2 + H_2O$

8. Complete the following word equations: (3 marks)

   a. acid + base → _____ + _____

   b. iron + oxygen → _____

   c. acid + carbonate → _____ + _____ + _____

9. Write the correct formula below all the substances in the equations below. Balance the equations. (6 marks)

   a. sulphuric acid + magnesium hydroxide → magnesium sulphate + water

   b. nitric acid + zinc → zinc nitrate + hydrogen

10. Below are two reactions that we performed in the classroom during this topic. For each of them, complete the word equation and write a balanced chemical equation. (8 marks)

    a. Word Equation:

       sulphuric acid + potassium hydroxide →

       Chemical Equation:

    b. Word Equation:

       hydrochloric acid + sodium carbonate →

       Chemical Equation: